

Practical - 4

Practical Title: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem Definition: Add dynamic UI components such as a notification banner, interactive buttons, FAQ/events controls, and a light/dark theme switcher using JavaScript. The StudentHub portal should become more interactive and easier to use.

Theory

1. How are DOM elements selected and modified?

Answer: JavaScript can select HTML elements using methods like `querySelector()` and `querySelectorAll()`. After selecting an element, its text, class, style, or other properties can be changed.

2. Are event listeners attached correctly?

Answer: Yes. Event listeners are used with buttons and other elements so that an action happens when the user clicks or interacts with them.

3. Is `localStorage` used for remembering theme choice?

Answer: A theme choice can be saved in `localStorage` so that the selected light or dark mode can be remembered when the page is opened again.

4. Does interactivity improve usability without breaking accessibility?

Answer: Yes. Simple buttons, clear messages, and a light/dark mode make the website easier to use. The controls should also have readable text and proper labels.

Work Done

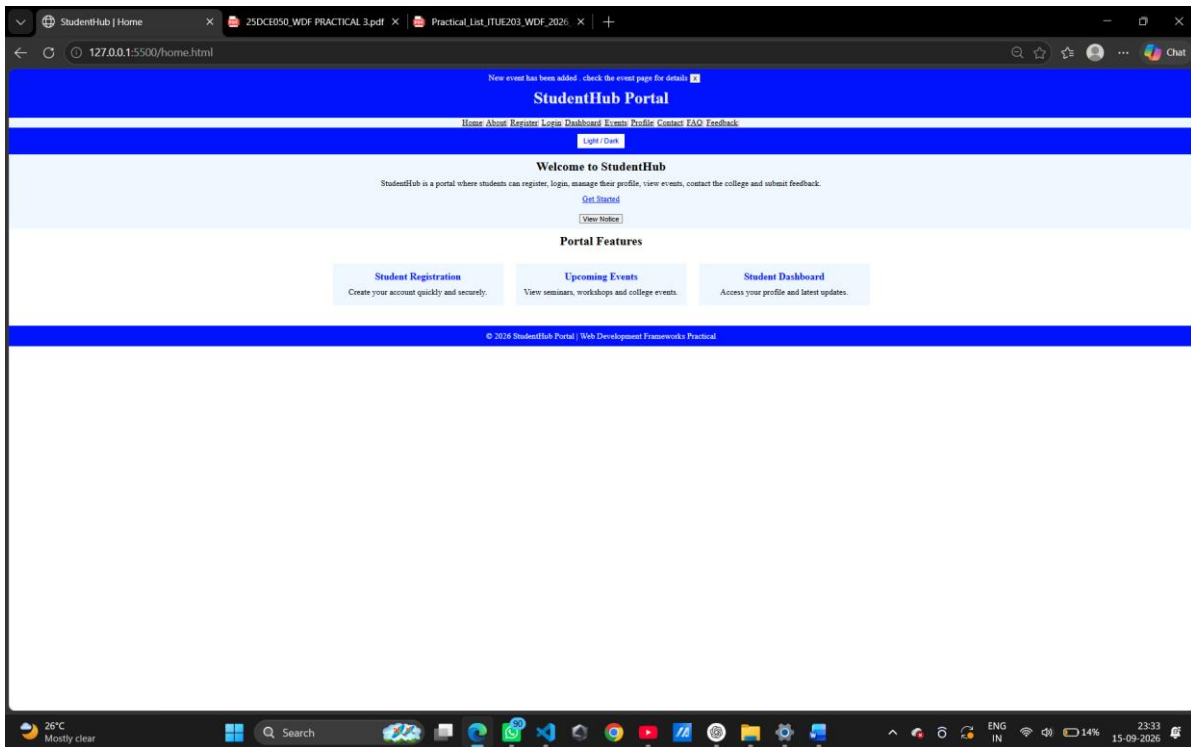
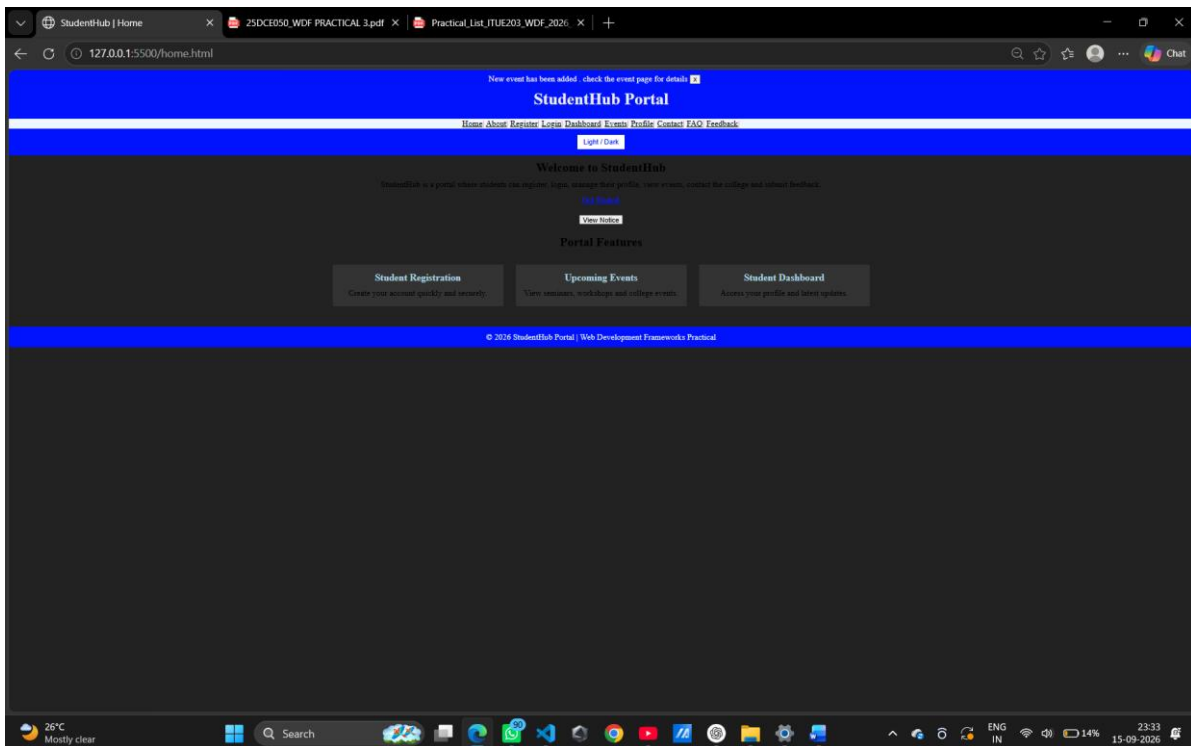
- Used JavaScript to handle button click events.
- Added interactive notification/message behaviour on the home page.
- Added a light and dark theme option for the StudentHub pages.
- Tested the interaction on different pages such as Home, About, Events, Register and FAQ.
- Used HTML, CSS and JavaScript together to make the static pages interactive.

Result

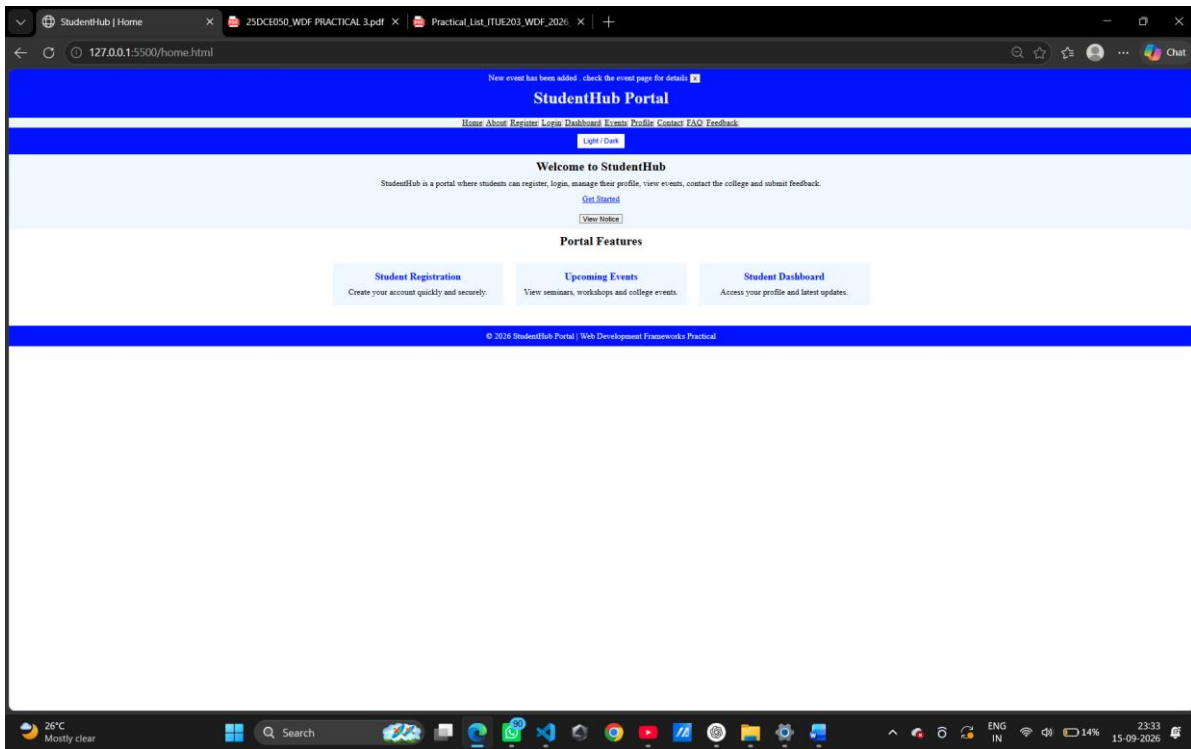
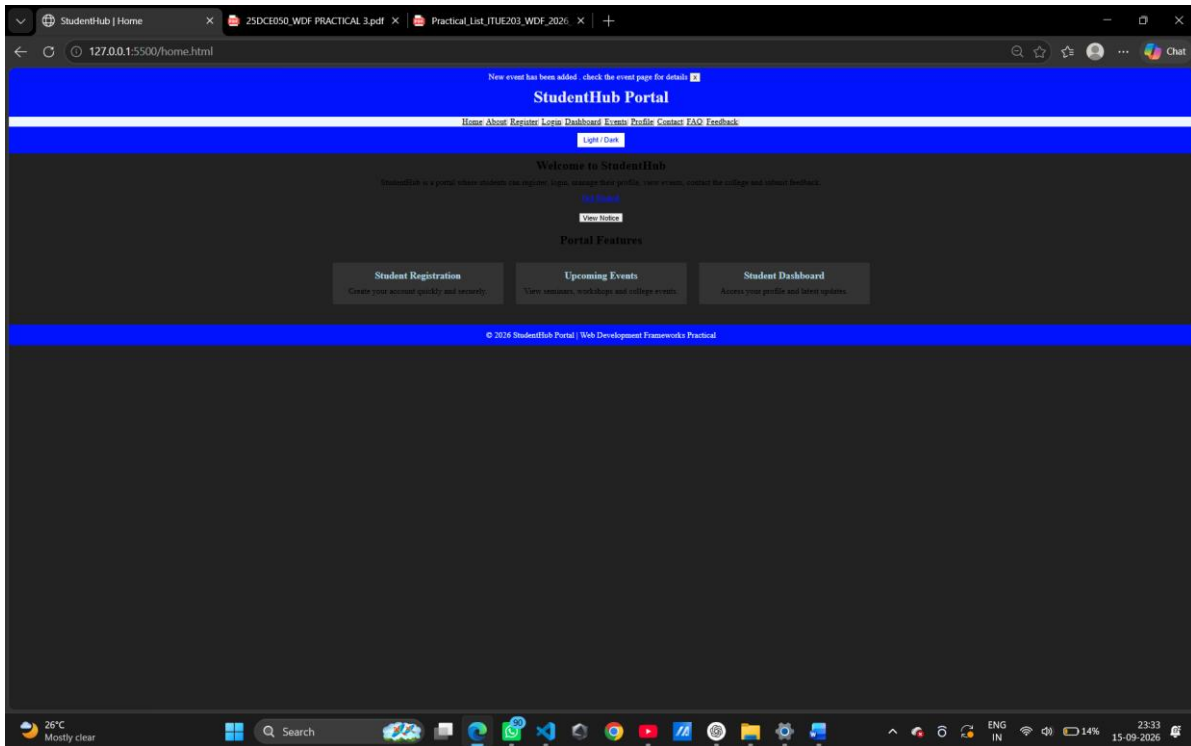
The StudentHub website became more interactive using JavaScript. Buttons respond to user actions, the notification can be interacted with, and the light/dark theme can be changed. The pages were tested in the browser and the required UI interactions were visible.

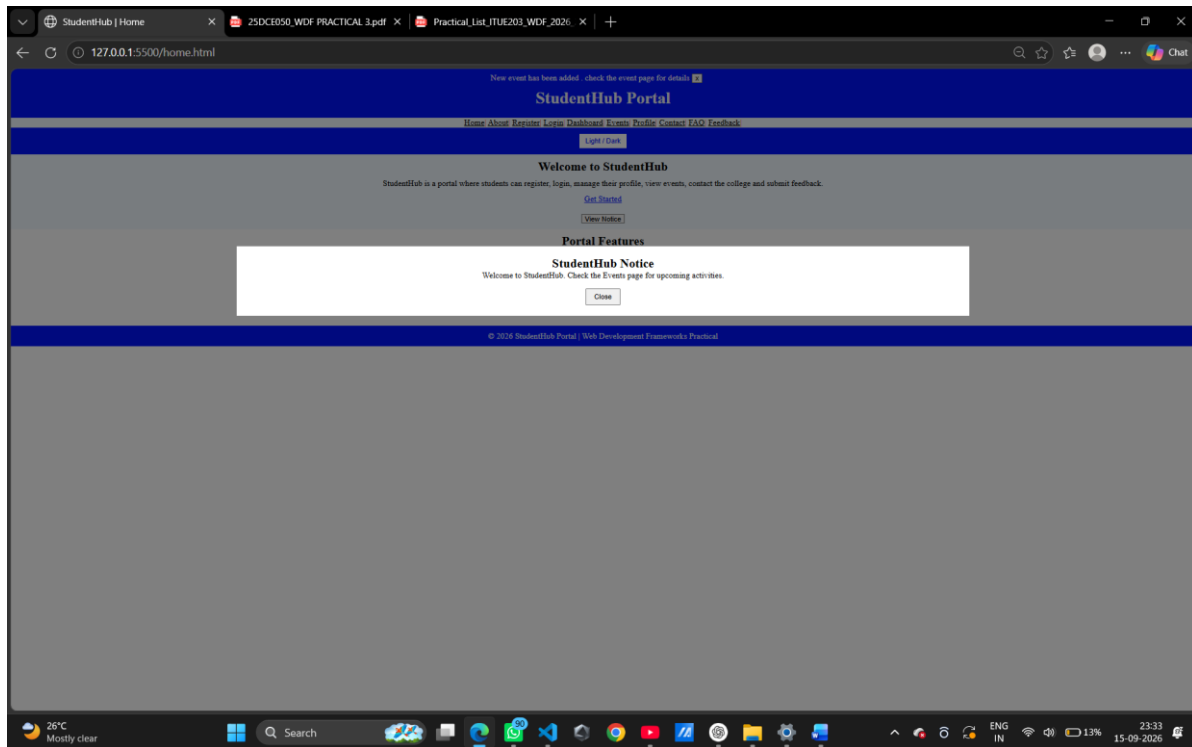
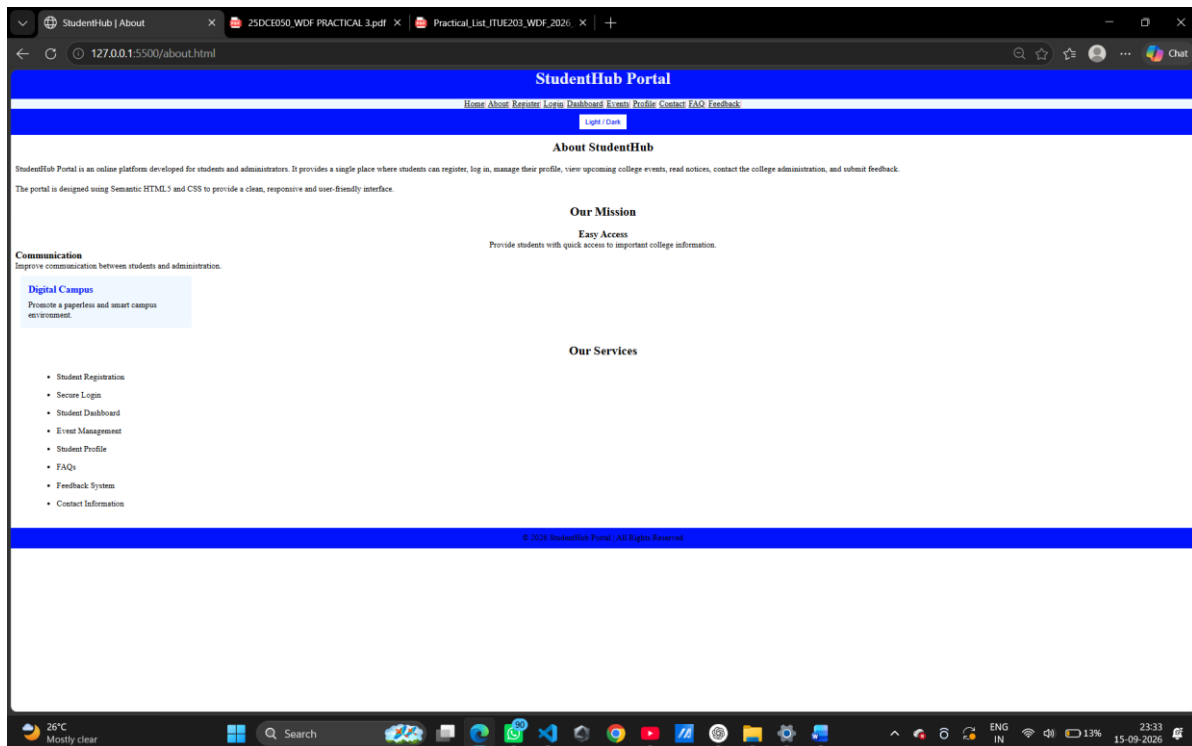
Evidence of Work (Screenshots)

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*Home page in light mode**Home page in dark mode*

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*Home page showing the light theme**Home page showing the dark theme*

*Notification popup/message**About page*

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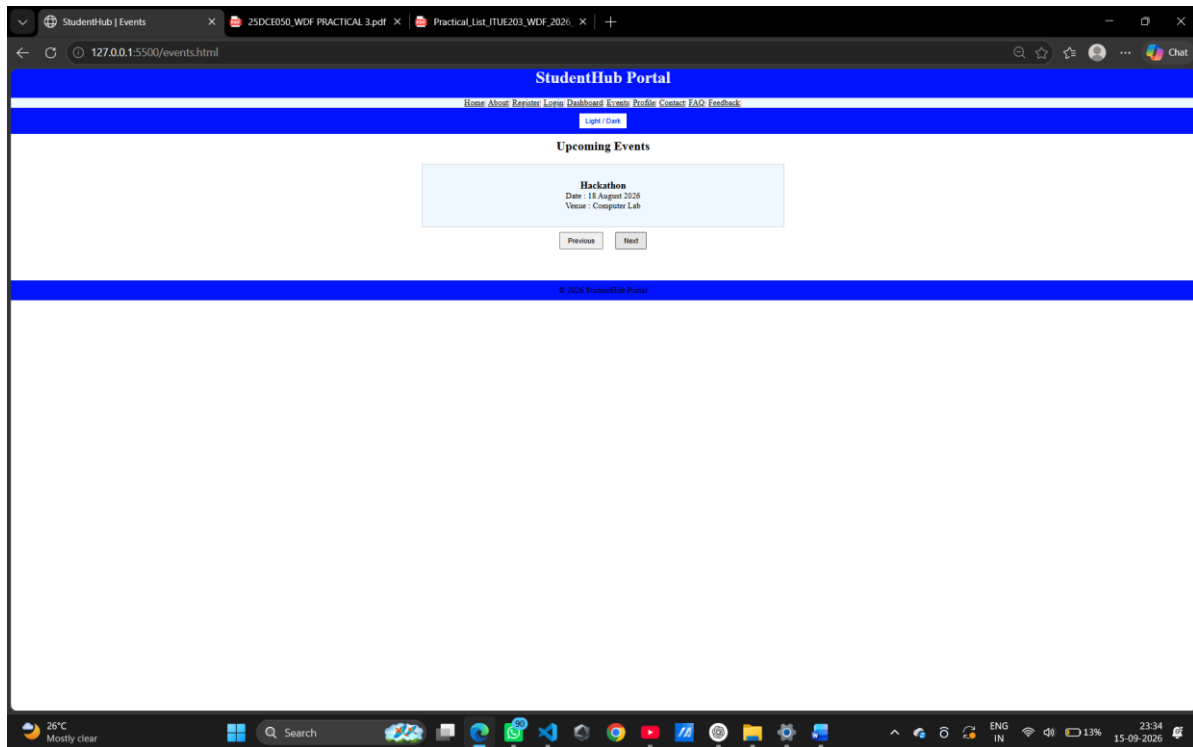
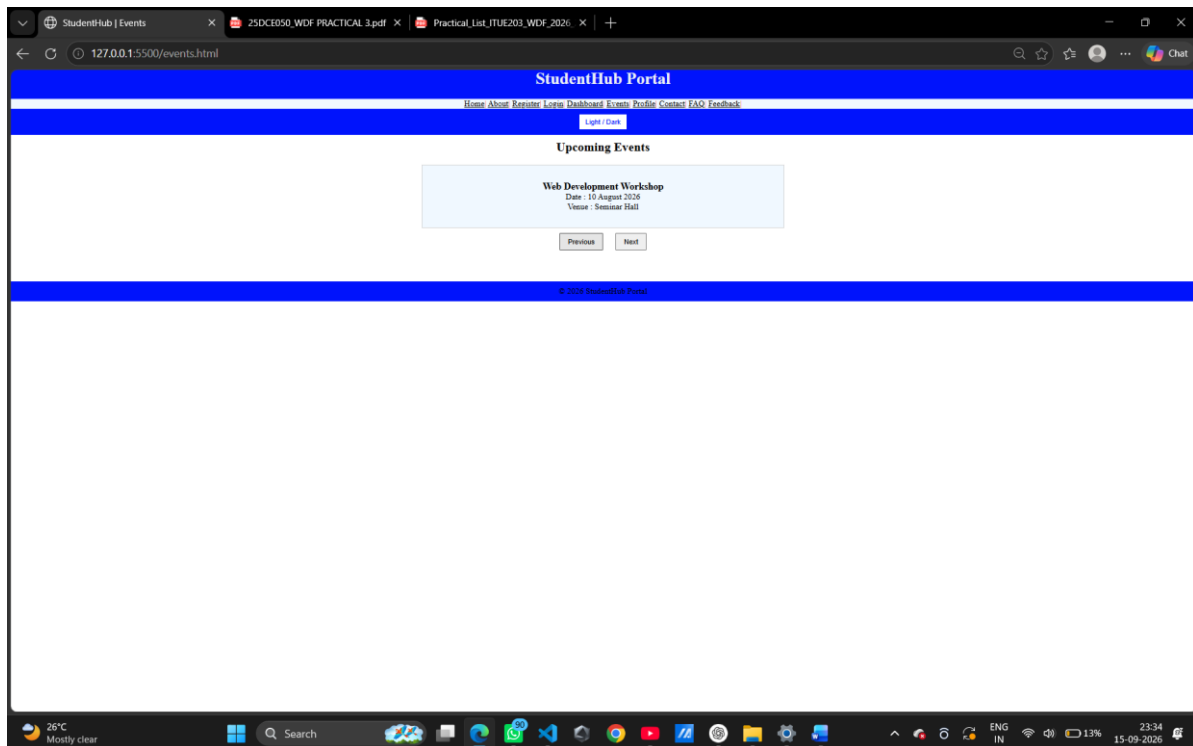
The screenshot shows a web browser window with the URL `127.0.0.1:5500/register.html`. The page has a blue header with the title "StudentHub Portal" and a navigation menu: [Home](#) [About](#) [Register](#) [Login](#) [Dashboard](#) [Events](#) [Profile](#) [Contact](#) [FAQ](#) [Feedback](#). A "Light / Dark" toggle is also present. The main content area is titled "Student Registration" and contains a form with the following fields: "Full Name" (text input), "Email Address" (text input), "Mobile Number" (text input), "Course" (dropdown menu), "Year" (dropdown menu), "Gender" (radio buttons for Male, Female, Other), "Password" (text input with a "Create Password" button), and "Confirm Password" (text input with a "Confirm Password" button). There is a checkbox for "I agree to the Terms & Conditions" and a "Register" button. The footer of the page reads "© 2025 StudentHub Portal | All Development Privately". The Windows taskbar at the bottom shows the date as 15-09-2025 and the time as 23:34.

Register page

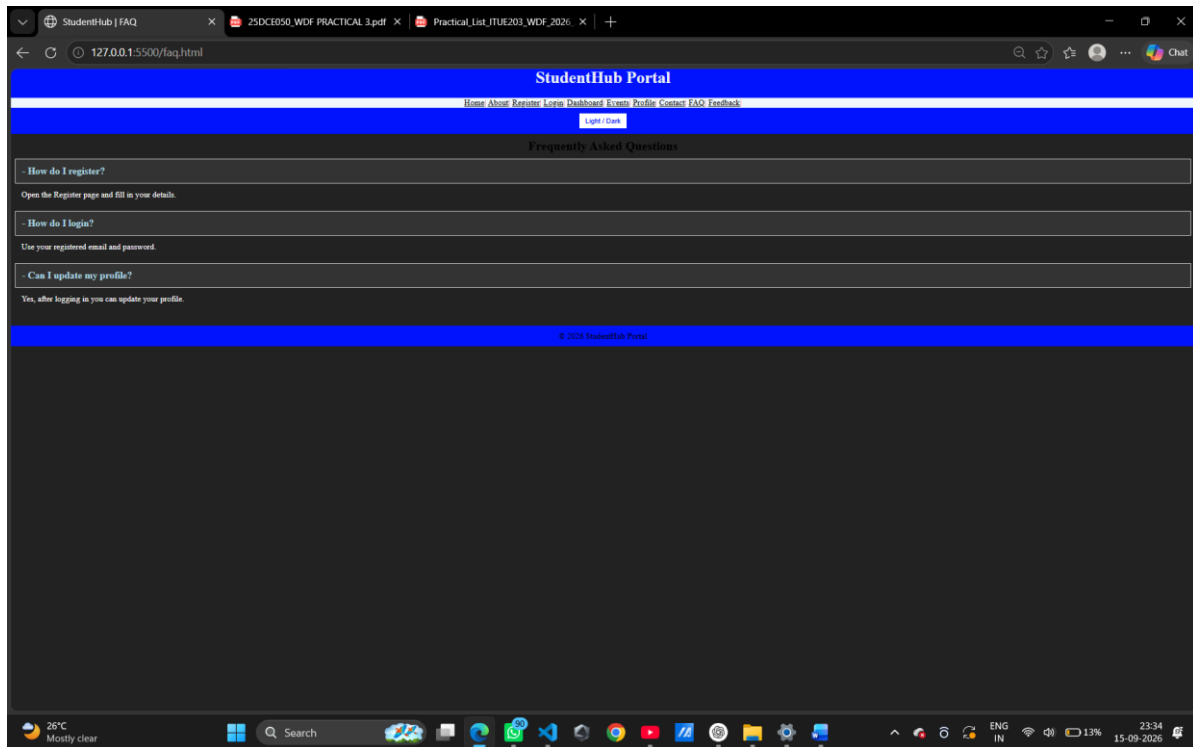
The screenshot shows a web browser window with the URL `127.0.0.1:5500/events.html`. The page has the same blue header and navigation menu as the register page. The main content area is titled "Upcoming Events" and features a light blue box containing the details of the first event: "Web Development Workshop", "Date : 10 August 2026", and "Venue : Seminar Hall". Below the event details are "Previous" and "Next" buttons. The footer of the page reads "© 2026 StudentHub Portal". The Windows taskbar at the bottom shows the date as 15-09-2025 and the time as 23:34.

Events page - first event

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*Events page - another event**Events page - event navigation*

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*FAQ page in dark mode*

GitHub Repository Link:

<https://github.com/vatsal1950/basics-of-html.git>

Challenges Faced:

I faced some difficulty in connecting JavaScript with the HTML elements and making the theme change work on the pages. I also had to check the buttons and page navigation again after adding the interactive features. Testing the website in the browser helped me find and correct these small problems.

Learning Outcomes:

- I learned how to use JavaScript to select and change HTML elements.
- I learned how to use click events to make buttons interactive.
- I learned how JavaScript can be used with HTML and CSS to improve a website.
- I learned how a light/dark theme can be changed using JavaScript and CSS.
- I learned how to test interactive web pages in a browser and fix simple errors.

Faculty Remarks: